2025W1 BIOL 300 Midterm

Instructions:

1. Make sure that your copy of the test includes 10 pages, including this one.
2. Please round final answers as specified within the question (if rounding is needed). We will mark final answers with a tolerance for intermediate rounding but aim to retain 4-5 decimal places in intermediate calculations.
3. Some questions have a box for the final answer. Please put the final answer in the box and show your work in the other space provided.
4. Answers can be in ink or pencil.
5. Other than a basic/scientific calculator, no electronic devices are permitted. Please turn off your cell phone and leave it in your bag.
6. Students who engage in dishonest practices will be subject to disciplinary action.
7. By taking this exam, you are declaring that your answers are all your own work.
8. Students taking the exam may not speak or in any other way communicate with other students or people outside the examination room.
9. Students may not expose their written paper to other students. The excuse of accidental exposure, forgetfulness, or ignorance will not be accepted.

Please fill out the box below. By signing, you acknowledge that you have read and fully understand the instructions above.

Student number								
Section								
Signature								
Name								

Name:_____ Student Number:_____

1. Use the set of measurements below (all in cm) for all questions on this page. Pay attention to the units, and remember to convert to the desired units (1 cm = 10 mm; 1 m = 100 cm).

   12, 18, 19, 12, 15, 23, 13

   a. (2 point) What is the mean of these data, in **cm**?
   (12 + 18 + 19 + 12 + 15 + 23 + 13) / 7 = 16

   Answer: 16

   b. (3 points) What is the standard deviation, in **cm**, rounded to four decimal places?

   4.1633319989 cm

   Answer: 4.1633

   c. (2 points) What is the median, in **mm**?
   Sort: 12, 12, 13, 15, 18, 19, 23
   median = 15 cm = 150 mm

   Answer: 150

   d. (2 points) What is the standard error of the mean, in **cm**, rounded to four decimal places?
   4.1633 / sqrt(7) = 1.5735915849 cm

   Answer: 1.5736

   e. (2 points) What is the approximate 95% confidence interval of the mean, in **cm**, using the rule of thumb that we learned in class? Round to four decimal places.
   16±2(1.5736)
   = 16 ± 3.1472

   Answer: (16 ± 3.1472) or (12.8528, 19.1472)

   f. (3 points) What is the variance, in **m^2**, rounded to four decimal places?

   Sample variance $s^2 = 17.333333\ cm^2$
   $1cm^2 = 10^{-4}\ m^2$
   $17.333333 * 10^{-4} = 0.0017333333$ m

   Answer: 0.0017

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2.

a. In a sample of 1923 healthy adults, mean fasting plasma glucose was found to be 93.4 mg/dL, with a 95% CI of 92.0 to 94.8 mg/dL.

(2 points) Assuming this variable is normally distributed, circle *all* that are true.

i. The standard error of the mean is approximately 1.4 mg/dL
ii. The standard error of the mean is approximately 0.7 mg/dL **Yes**
iii. The standard deviation of the data is approximately 1.4 mg/dL
iv. The standard deviation of the data is approximately 0.7 mg/dL

b. (2 point) Which of the following is the most appropriate interpretation of the 95% CI given in Question 2a? Circle one.

i. There is a 95% probability that the true population mean is in the interval 92.0 to 94.8 mg/dL (inclusive)
ii. We are 95% confident that the true population mean is in the interval 92.0 to 94.8 mg/dL (inclusive) **Yes**
iii. 95% of the raw data are within the interval 92.0 to 94.8 mg/dL (inclusive)
iv. Trick question, all other statements are wrong.

c. (3 points) The following block of R code is intended to compute the standard error of the mean of a vector called `vec` in a fresh RStudio session. Please identify and explain any issues with the code that would result in an incorrect calculation. If there are none, write "No errors".

```
vec <- c(7, 102, 98.6, 8, 85, 64, 54)
n <- length(vec)
sd(vec) / (n)
```

Should divide by square root of n, not n

d. (3 points) The following block of R code is intended to compute the coefficient of variation for a variable called `age` within the file `titanic.csv` in a fresh RStudio session. Please identify and explain any issues with the code that would result in an incorrect calculation. If there are none, write "No errors".

```
read.csv("DataForLabs/titanic.csv", stringsAsFactors = TRUE)
100*sd(titanicData$age, na.rm = TRUE)/mean(titanicData$age, na.rm = TRUE)
```

Csv is read in but not saved to environment; titanicData would not exist

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3.

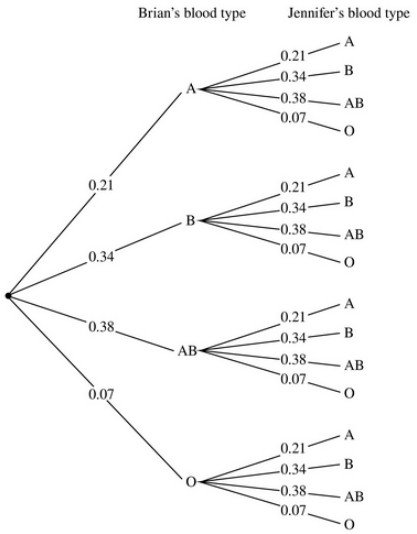
a. (5 points) A flycatcher is trying to catch passing bugs. The probability that it catches a bug on any given try is 20%. What is the probability that it takes at least four tries to catch its *first* bug? Give your answer in decimal form, rounded to three decimal places.

Pr[*miss*]=1−Pr[*success*]=1−0.2=0.8

Pr[*first success on the fourth try or later*]=0.8×0.8×0.8=0.512

Answer: 0.512

b. (5 points) A recent study shows that 21% of adults have blood type A, 34% of adults have blood type B, 38% of adults have blood type AB, and 7% of adults have blood type O. What is the probability that two people chosen at random share the same blood type? Give your answer as a decimal, rounded to three decimal places.



Pr[*same type*]= $0.21^2 + 0.34^2 + 0.38^2 + 0.07^2$
$=0.0441+0.1156+0.1444+0.0049=0.3090$

Answer: 0.309

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4.

A data frame called `iris` exists in your R environment. You run:

```
library(dplyr)
summary(iris)
```

Which results in:

```
> summary(iris)
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width          Species
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa     :50
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50
 Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
```

You next run the line:

```
filter(iris, Species == "setosa")
```

Mark each statement below about the outcome of running this line of code as True or False in the column on the right (2 points each).

	TRUE or FALSE?
The line will fail because using "==" is not correct.	**FALSE**
The result will have 50 rows.	**TRUE**
The result will have 100 rows.	**FALSE**
The data frame `iris` is overwritten by a new data frame.	**FALSE**
A new data frame is printed to the Console.	**TRUE**
You see a new data frame with only the rows for which the `Species` was not `setosa`.	**FALSE**

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5. Many animal species produce offspring with biased sex ratios, meaning male and female offspring are not equally likely. Scientists randomly sampled 10 Magellanic penguin chicks, and observed that 8 of them were male, and 2 were female.

   a. (12 points) Based on this sample, is there evidence of a biased sex ratio in this population of Magellanic penguins? Conduct an appropriate statistical test. Assume $\alpha = 0.05$. Clearly state null and alternative hypotheses and the name of the test you are using. Give the *P-value* as precisely as possible and interpret your result in words. Assume random sampling.

H0: The proportion of females, p, equals 0.5
HA: The proportion of females, p, does not equal 0.5

If a Chi-squared GoF test is attempted, you should see that Expected values are each 5, which violates the assumption of the test. Therefore, performing a Chi-squared GoF would be invalid.

The test to use is the Binomial test:

Calc:

$$P = 2\left(\binom{10}{0}(0.5)^0(0.5)^{10} + \binom{10}{1}(0.5)^1(0.5)^9 + \binom{10}{2}(0.5)^2(0.5)^8\right)$$

$$= 2(1(0.5)^{10} + 10(0.5)^{10} + 45(0.5)^{10})$$

$$= 2\left(\frac{1 + 10 + 45}{1024}\right) = 2(0.0546875) = 0.109375$$

Therefore, P = 0.1094 > 0.05

Fail to reject H_0. With this sample, there isn't sufficient evidence of a biased sex ratio in this population.

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b. (4 points) Calculate the 95% confidence interval for the proportion of female chicks in the entire population, described on the previous page. Give your answer in decimal form, rounded to four decimal places.

$$p' = \frac{4}{14} = 0.285714$$

$$\textbf{Lower bound}: 0.285714 - 1.96\sqrt{\frac{0.285714(1-0.285714)}{14}} = 0.049073$$

$$\textbf{Upper bound}: 0.285714 + 1.96\sqrt{\frac{0.285714(1-0.285714)}{14}} = 0.522355$$

Answer: 0.0491 to 0.5224

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6. The BAFTA study was a randomized trial in the UK that compared two treatments for preventing strokes in people over 75 with atrial fibrillation. Participants either received warfarin (a blood thinner that requires regular blood tests) or aspirin (a common over-the-counter drug). One safety concern was the risk of a dangerous side effect: intracranial hemorrhage. During the study, researchers recorded the results in the table below. They were interested in the question: does the risk of intracranial hemorrhage differ between patients given warfarin vs. aspirin?

Group	Intracranial hemorrhage	No intracranial hemorrhage
Warfarin	2	486
Aspirin	1	484

a. (2 points) State the null and alternative hypotheses
H0: Treatment and risk are independent; $p_W = p_A$
HA: p_W not equal to p_A

b. (6 points) Fill the table below with expected values, rounded to 4 decimal places, based on H_0

Group	Intracranial hemorrhage	No intracranial hemorrhage	Total
Warfarin	**1.5046**	**486.4954**	**488**
Aspirin	**1.4954**	**483.5046**	**485**
Total	**3**	**970**	**973**

c. (3 points) Name the statistical test that you would use. Briefly why it is an appropriate choice (1-2 sentences). You do not need to perform the test itself.
Fisher's exact test, because some of the Expected < 5

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7.

a. (2 point) In a study of bird nesting site preferences (among tree cavities, cliff ledges, ground burrows, and artificial boxes), you expected birds to use all four nest types equally. Name the statistical test that you would use to test the null hypothesis that birds exhibit no preference for nest types.

Chi-squared goodness-of-fit test

b. (3 points) Based on the observed frequencies for the scenario in Question 7a, you calculate the value of the appropriate test statistic from your sample as 7.94. What is the associated P-value, as precisely as you can give it based on the statistical tables?

Based on the table, chisq 7.94 with 3 DoF would land between 0.05 and 0.025
Therefore 0.05 > P > 0.025

c. As you **<u>decrease</u>** the size of your sample from a given population, what will generally happen to the following? *For each of the blanks, please write one of:* ***increase, no change****, or* ***decrease****.* (2 points each)

i. Sample standard deviation: **No change**

ii. Standard error of the mean, estimated from sample ($SE_{\bar{Y}}$): **Increase**

iii. True standard error of the mean (population parameter $\sigma_{\bar{Y}}$): **Increase**

iv. Probability of Type I error: **No change**

v. Probability of Type II error: **Increase**

vi. Precision of an estimate: **Decrease**

vii. Sample range of a continuous variable: **Decrease**

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8.

a. (2 points) Can variance ever be negative? Please state yes or no and briefly explain why (1-2 sentences).
No, squaring deviations in the numerator will positivize the result

b. (6 points) Imagine that a forester is surveying Douglas fir trees. Trees with a DBH (diameter at breast height) between 37.4 cm and 67.5 cm are suitable for timber harvest. Assuming that the mean DBH of Douglas fir is 52 cm, with a variance of 225 cm^2, what is the probability that a Douglas fir tree, selected at random, will be suitable for harvest? Give your answer in decimal form, rounded to five decimal places.

$\mu = 52;\ \sigma = 15cm$

$z_1 = \frac{37.4 - 52}{15} = -0.9733$

$z_2 = \frac{67.5 - 52}{15} = 1.0333$

Based on Z dist table:
0.9733 -> 0.16602. The area to the left of -0.9733 is equal to 0.16602 by symmetry.
1.0333 -> 0.15151, but take 1 – value to get 0.84849
Therefore, area of interest: 0. 84849 – 0.16602 = 0.68247

Answer: 0.68247